

FINAL YEAR MAJOR PROJECT

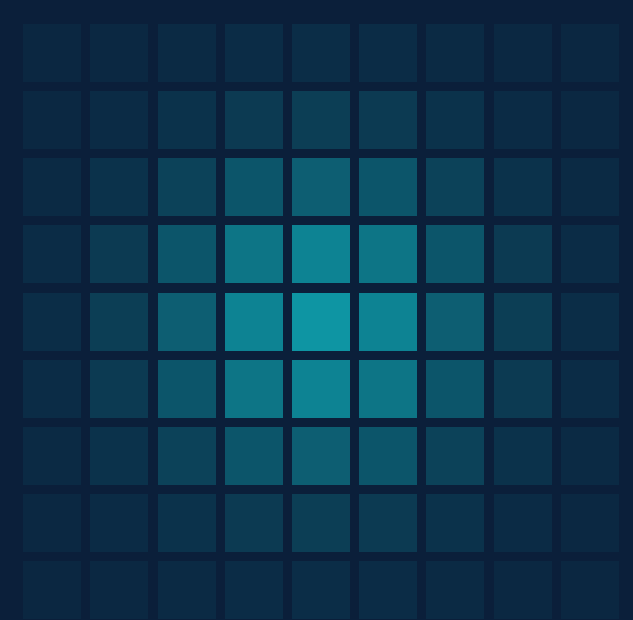
Deepfake Detection via Hybrid Spatial-Temporal-Frequency Feature Learning with Transformer-Based Fusion

Ayush Tiwari · Ayushman Patel · Atharv Shukla · Bhanupratap Chourasia

Department of _____ | _____ (Institution)

Faculty Guide: _____

Presentation Date: 12 September 2026



spatial · temporal · frequency

Project Team

Final Year Major Project · Deepfake Detection via Hybrid Spatial-Temporal-Frequency Feature Learning with Transformer-Based Fusion

AT

Ayush Tiwari

Enrollment No.: _____

AP

Ayushman Patel

Enrollment No.: _____

AS

Atharv Shukla

Enrollment No.: _____

BC

Bhanupratap Chourasia

Enrollment No.: _____

Department of _____ | _____ (Institution) | Faculty Guide: _____

Problem Identification

Why detection is hard, and why it is still an open problem

- Deepfake technology enables highly realistic manipulation of facial content.
- Human visual inspection is increasingly unreliable for judging authenticity.
- Modern generation techniques reduce the obvious visual artifacts that early detectors relied on.
- Detection models can become dependent on dataset-specific artifacts rather than general forgery cues.
- A detector trained on one manipulation distribution may perform poorly on unseen data: new generators, different compression, lighting or capture conditions.
- **Robust and generalizable detection therefore remains an important research problem.**

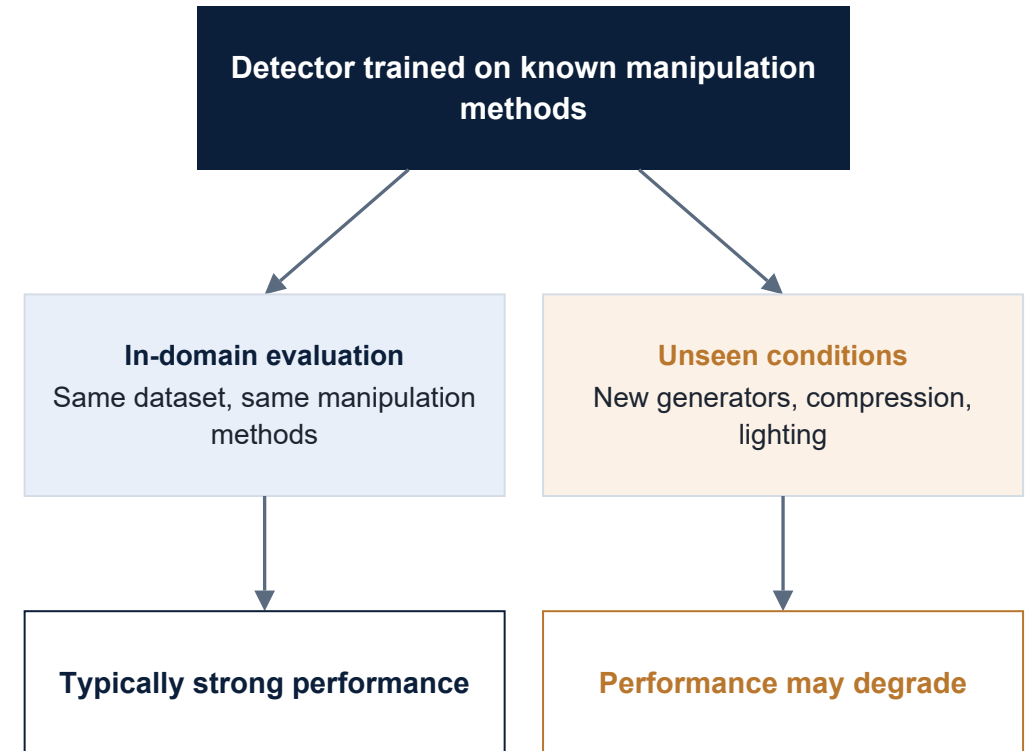
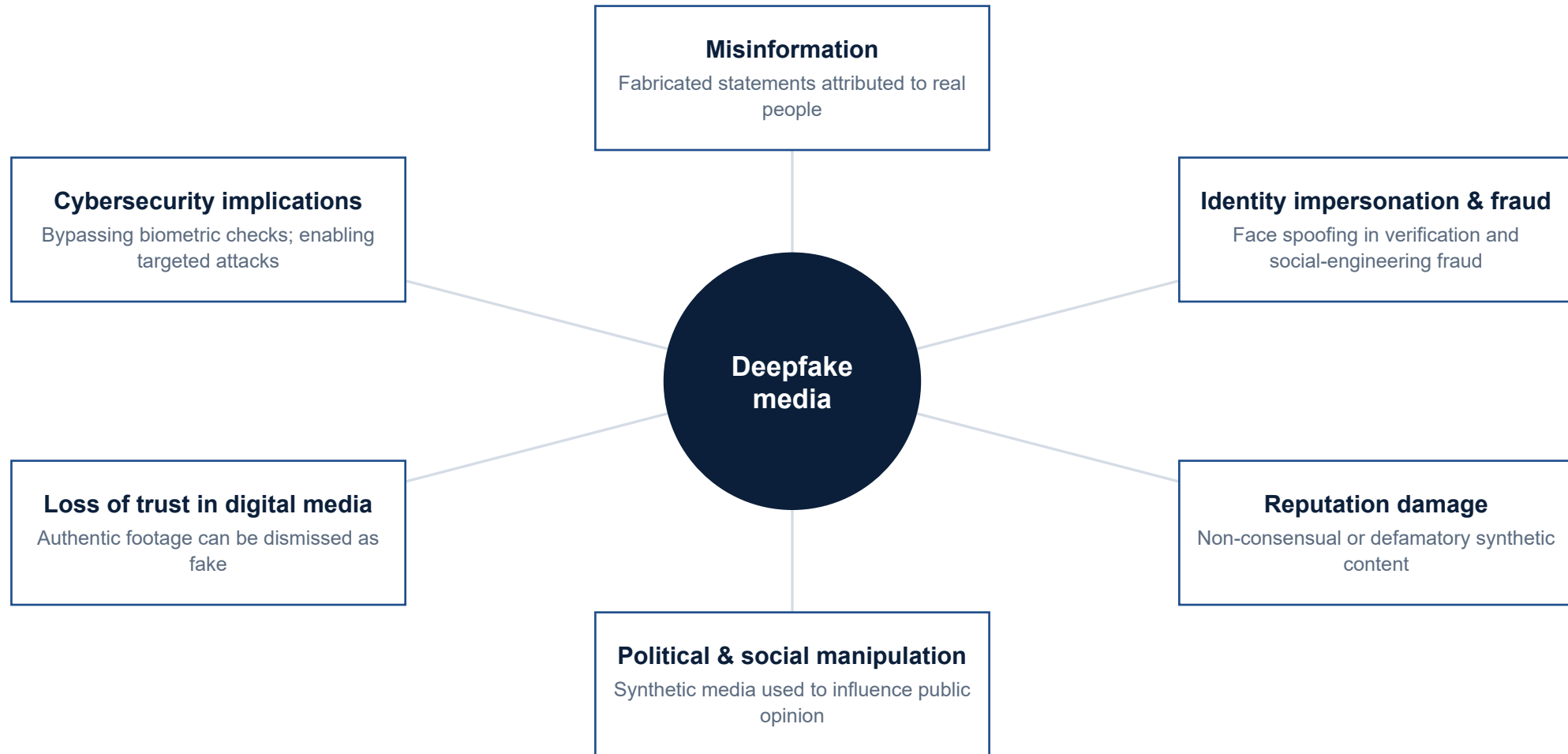


Illustration of the generalization problem this project targets. Not an experimental result.

Why Deepfake Detection Matters

Reliable detection is one component of a broader response to synthetic media. Consequences of undetected manipulation:



Detection Signals in Deepfake Media

SPATIAL

“What does the face look like?”

Individual-frame visual information

May reveal:

- Facial texture and edges
- Blending boundaries and artifacts
- Unnatural facial details
- Local visual inconsistencies

Captured by (proposed):

Pretrained CNN / vision backbone applied to each face frame

TEMPORAL

“How does the face change across frames?”

Relationships across the frame sequence

May reveal:

- Unnatural or jittery motion
- Frame-to-frame inconsistencies
- Blinking and facial movement patterns
- Temporal artifacts

Captured by (proposed):

Transformer encoder over the sequence of frame-level features

FREQUENCY

“What hidden pixel-level patterns exist beneath the visual appearance?”

Frequency-domain representation of image information

May reveal:

- Abnormal spectral distributions
- Up-sampling and blending traces
- Periodic patterns not visible in pixel space
- Attenuated high-frequency detail

Captured by (proposed):

FFT/DCT transform followed by a neural feature extractor

The three signals are complementary. The proposed framework is designed to combine them rather than rely on any single one; no single cue is assumed to be present in every deepfake.

Related Work

Landscape of prior approaches, grouped by the type of evidence they exploit. Bracketed numbers refer to the References slides.

Category	Main idea	Strength	Limitation	Representative work
CNN-based detection	Train a convolutional network on face crops to classify real vs. manipulated frames.	Strong in-domain accuracy; simple to train from pretrained backbones.	Tends to learn dataset-specific spatial artifacts; accuracy can drop on unseen manipulations or compression.	MesoNet [4]; CNN baselines in FaceForensics++ [1]
Vision Transformer / Transformer-based	Apply self-attention over image patches or CNN feature maps to capture global context.	Models long-range dependencies; flexible attention over regions.	Data-hungry; remains frame-centric unless explicitly extended to video.	ViT [6]; CNN + ViT for video deepfakes [16]
Temporal / video-based	Model frame sequences with recurrent or attention modules to find inconsistencies over time.	Uses motion and coherence cues that single frames cannot provide.	Higher compute; sensitive to frame sampling and face-tracking quality.	RNN-based [11], [12]; temporal coherence [13]; eye blinking [14]
Frequency-domain	Analyse spectra (FFT/DCT) to expose up-sampling and blending traces left by generators.	Sensitive to generator artifacts that are subtle in pixel space.	Artifacts can be attenuated by compression; not guaranteed for every method.	F3-Net [8]; Frank et al. [10]; Durall et al. [9]
Hybrid / multi-stream	Combine several cue types (spatial, frequency, temporal, attention maps) in one model.	Complementary evidence can improve robustness.	More complex; the fusion strategy and its benefit must be validated by ablation.	Multi-attentional [17]; Face X-ray [15]

Research Gap

Limitations observed across existing approaches

■ Dependence on spatial artifacts

Single-frame appearance cues are generator-specific and easy to over-fit.

■ Limited modeling of temporal consistency

Many detectors treat video as independent frames.

■ Under-utilization of frequency-domain information

Spectral cues are often ignored or used in isolation.

■ Dataset-specific bias

Benchmarks share capture conditions and generators; models absorb these regularities.

■ Weak cross-dataset generalization

Performance frequently drops when evaluated on a different dataset.

RESEARCH QUESTION

Can combining spatial, temporal, and frequency-domain information through attention-based feature fusion improve the robustness of deepfake detection?

HYPOTHESIS (to be tested, not yet validated)

Complementary cues, fused with attention, may reduce dependence on any single artifact type and therefore hold up better under distribution shift (new manipulation methods, compression, capture conditions).

Proposed Solution

PROPOSED

01

Spatial branch

Extract spatial information from RGB face frames with a pretrained CNN / vision backbone.

02

Frequency branch

Transform face frames into frequency representations (FFT/DCT) and learn discriminative frequency features.

03

Temporal modeling

Learn relationships across the sampled frame sequence with a Transformer encoder.

04

Attention-based fusion

Combine the complementary representations using Transformer-based attention rather than fixed weighting.

05

Binary classification

Classification head producing a REAL / FAKE decision with a confidence score.

06

Evaluation protocol

Measure in-domain performance and, where feasible, cross-dataset generalization; ablate each component.

Project status

CONFIRMED

Topic and research direction

PROPOSED

Hybrid architecture (this deck)

PLANNED

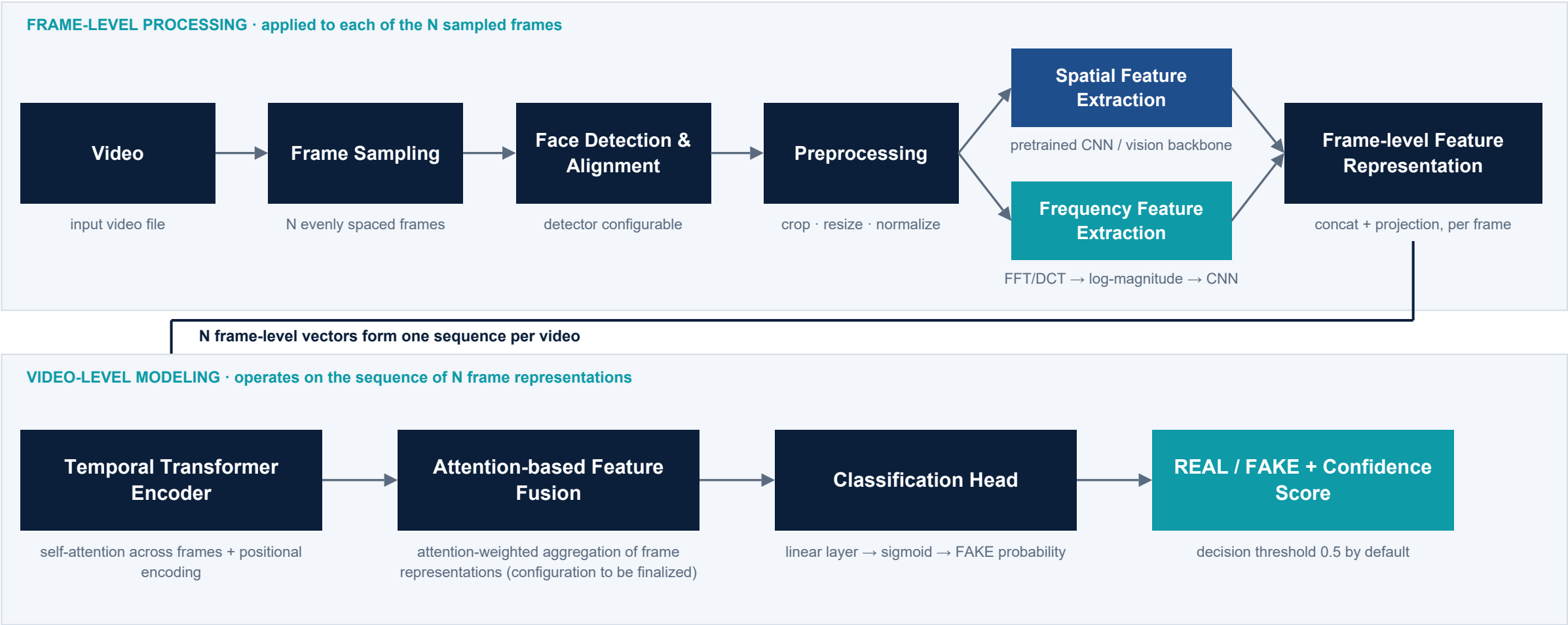
Preprocessing, training, experiments

NOT YET AVAILABLE

Experimental results

Methodology

PROPOSED

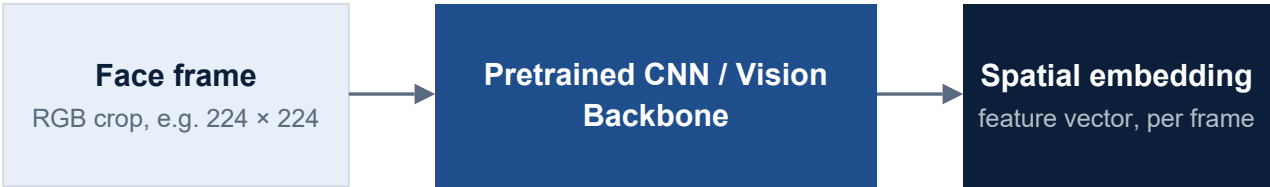


The diagram reflects the proposed design. The exact backbone, face detector and fusion configuration will be selected during implementation.

Spatial Feature Extraction

PROPOSED

Face Frame → CNN → Spatial Embedding



Backbone: final architecture to be selected during implementation. An ImageNet-pretrained ResNet18 is the initial candidate in the project implementation plan, kept configurable.

Input	One face frame (RGB image)
Processing	CNN / vision backbone, pretrained on natural images; classification layer removed
Output	Fixed-length spatial embedding describing the visual characteristics of the frame

Examples of information the embedding may capture

- Texture of skin and hair regions
- Edges and contour sharpness
- Facial structure and symmetry
- Blending artifacts at face boundaries
- Local inconsistencies in colour and lighting

Why spatial information alone may be insufficient

- Spatial artifacts are generator- and dataset-specific: a model can over-fit to them and fail on unseen manipulation methods.
- Compression, resizing and post-processing weaken fine-grained pixel cues.
- A single frame carries no information about motion or consistency over time.

Frequency-Domain Feature Extraction

PROPOSED

Concept

- An image can be represented in the spatial domain (pixels) or in the frequency domain.
- The frequency representation describes how rapidly pixel intensities vary across the image.
- Low-frequency components represent smooth structures; high-frequency components capture fine detail and rapid changes.
- Deepfake generation and post-processing (up-sampling, blending, re-compression) **may** introduce abnormal frequency patterns.
- An FFT or DCT transforms a face frame into a frequency representation (e.g. a log-magnitude spectrum).
- A neural network can then learn which frequency patterns are discriminative.

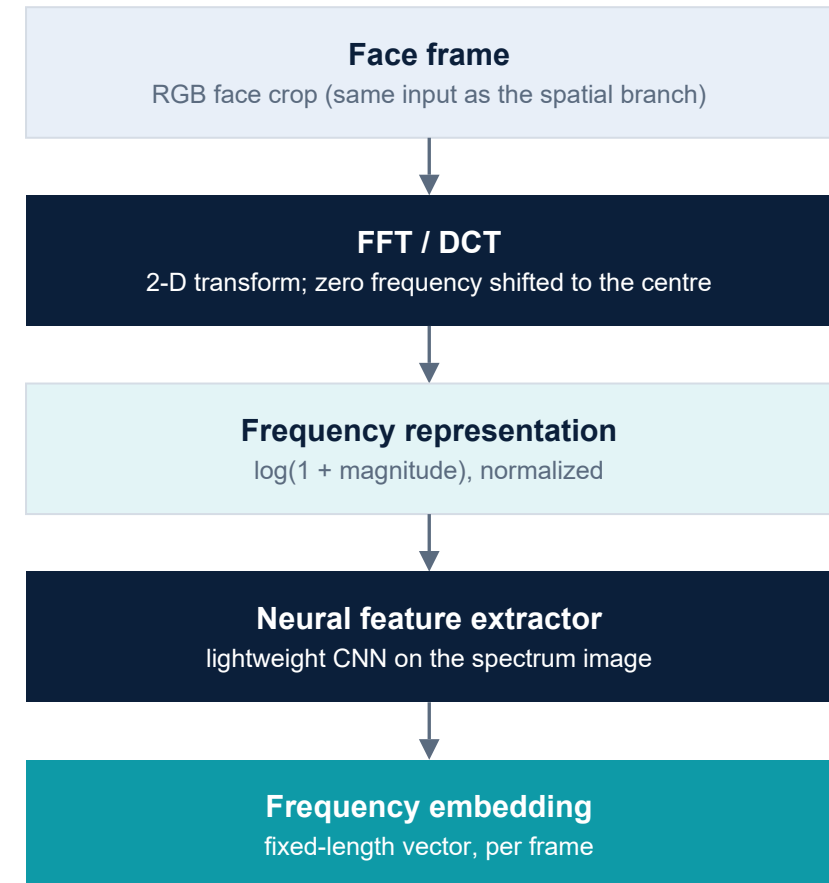
Low frequency

Smooth regions, overall shape and shading

High frequency

Edges, texture, noise, up-sampling traces

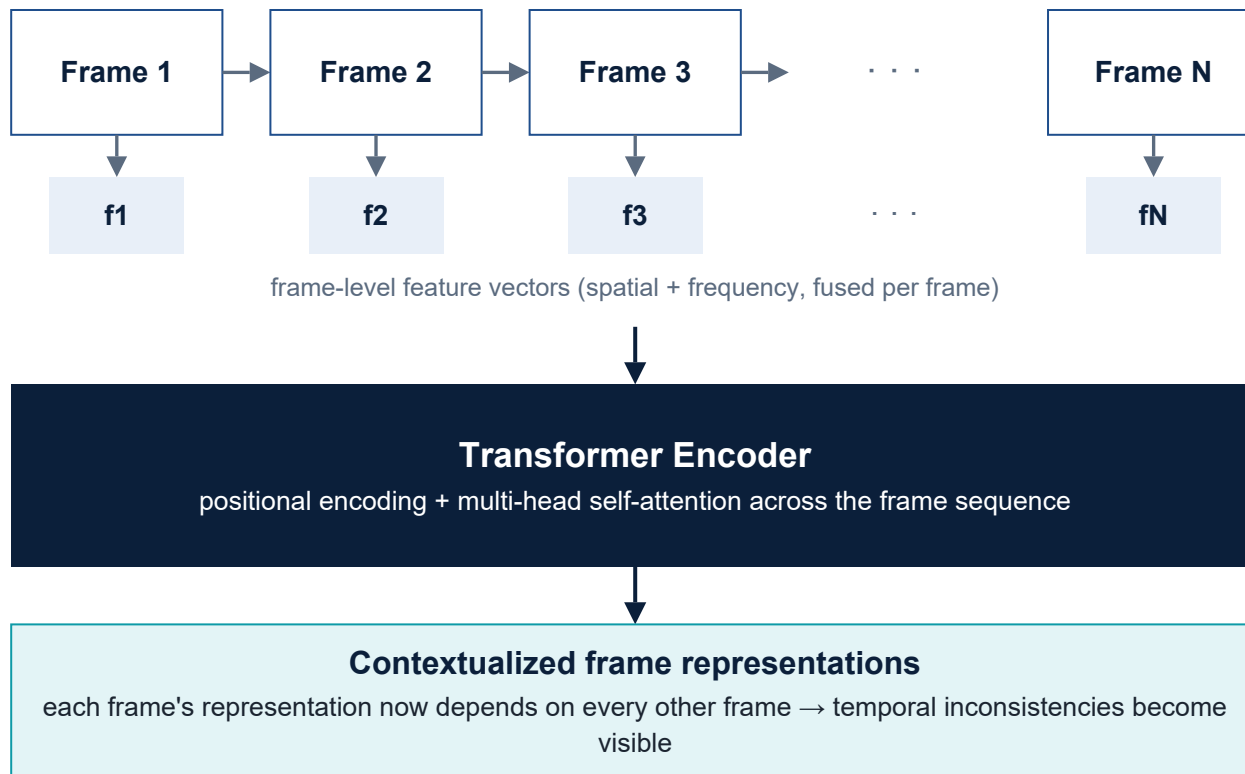
Frequency cues can provide additional discriminative information, but they are not assumed to be present in every deepfake.



Temporal Feature Modeling

PROPOSED

A video is a sequence of frames; each frame yields a feature vector



Each element attends to all others, so the model can relate distant frames directly.

Why a Transformer encoder rather than a recurrent model (proposed)

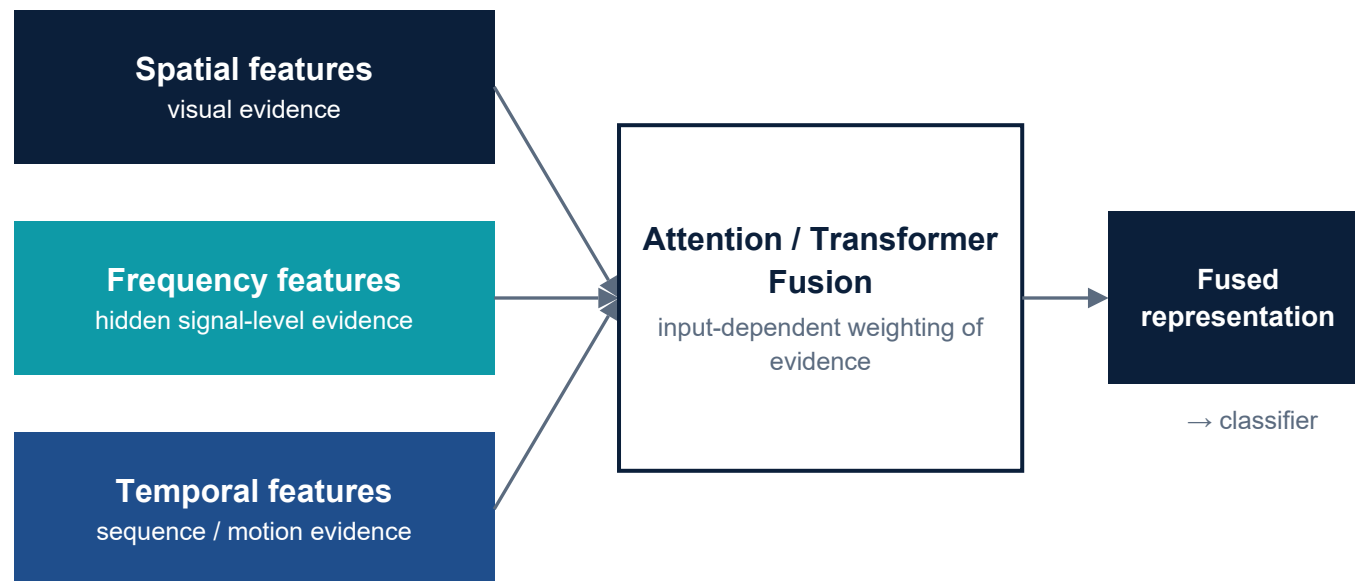
- Self-attention models relationships between distant frames directly, without passing information step by step.
- Sequence elements are processed in parallel during training.
- It provides a flexible mechanism for learning temporal dependencies and for attention-based fusion in the same framework.

NOT A LANGUAGE MODEL

Transformer ≠ LLM. The encoder used here has no tokenizer, no vocabulary and no language objective. It is a standard sequence model (e.g. PyTorch's Transformer encoder) applied to numeric frame embeddings, as in vision and video transformers [6], [7].

Attention-Based Feature Fusion

PROPOSED



PROPOSED MECHANISM

Spatial and frequency embeddings are fused per frame; the Transformer encoder's self-attention then lets frame representations interact across the sequence; a video-level representation is obtained by aggregation (mean pooling or attention-weighted, to be finalized during implementation).

Why attention rather than fixed weighting

Attention allows the model to learn which information source is more relevant for a particular input, instead of assigning identical importance to every feature source.

Intuition: an illustrative example (not a result)

Evidence source	Observation for this input	Attention weight
Spatial	appearance mostly normal	lower
Temporal	suspicious motion pattern	higher
Frequency	suspicious spectral pattern	higher

For this example the fusion mechanism can learn that the temporal and frequency evidence is more informative, and rely on it more than on the spatial evidence.

Dataset & Preprocessing Strategy

PLANNED

FaceForensics++ [1]

PRIMARY · training and in-domain evaluation

Standard benchmark for deepfake detection research: real videos plus several manipulation methods, enabling reproducible comparison.

Celeb-DF [2]

GENERALIZATION · cross-dataset evaluation, if feasible

Different deepfake characteristics from the training data; used only for testing (no fine-tuning) to probe generalization.

DFDC [3]

OPTIONAL · subject to available compute and time

Very large and computationally expensive to preprocess and evaluate; considered only after the above are complete.

Preprocessing pipeline (per video)

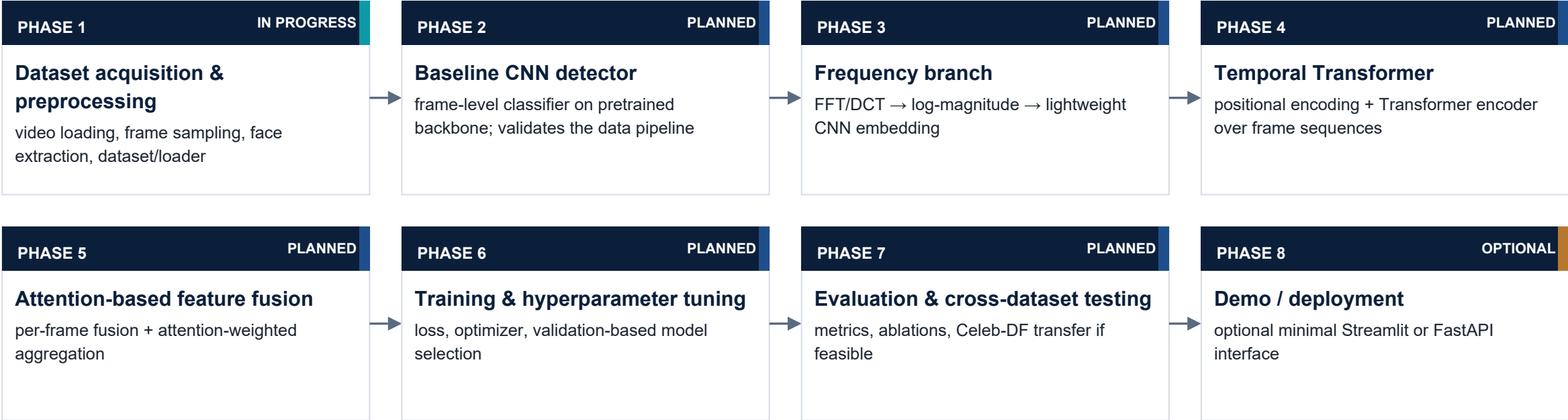
- 1 Video loading
- 2 Frame sampling (N frames, evenly spaced)
- 3 Face detection
- 4 Face alignment / cropping
- 5 Resize to a fixed resolution
- 6 Normalization
- 7 Sequence construction (N face crops in order)
- 8 Label assignment

Labels: REAL = 0, FAKE = 1 (assigned per video)

Frames inherit the label of their video; the model predicts at video level from the frame sequence. Train / validation / test splits are formed at video level so frames of one video never appear in two splits.

Implementation Plan

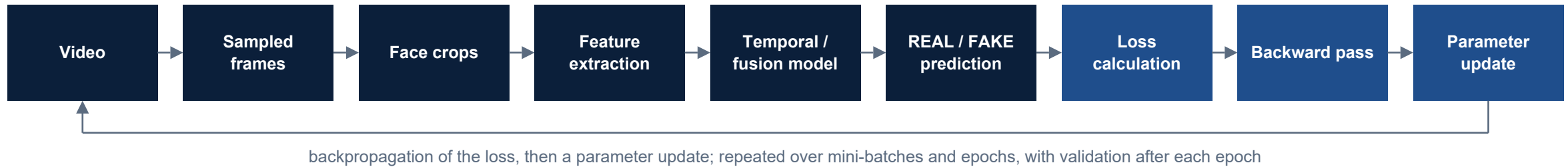
PLANNED



Core stack confirmed	Python	PyTorch	OpenCV	NumPy	SciPy (where required)	scikit-learn
Optional tools only if actually used	Streamlit / FastAPI (demo interface)		TensorBoard / Weights & Biases (tracking)		Face detector library (configurable)	

Training & Experimental Design

PLANNED



Training components (planned)

- Binary cross-entropy loss on the video-level logit (REAL = 0, FAKE = 1)
- Adam / AdamW optimizer
- Transfer learning: pretrained backbone for the spatial branch where appropriate
- Held-out validation set for model selection and hyperparameter choices
- Early stopping / checkpointing of the best validation model, if implemented
- Modest augmentation (flip, small rotation, mild colour jitter) applied before the frequency transform

Experiment design: Baseline vs. Proposed

BASELINE

CNN-based detector

Frame-level classification; per-frame outputs averaged over the sampled frames. No frequency branch, no temporal model.

PROPOSED

Hybrid model

Spatial + frequency + temporal Transformer with attention-based fusion; one prediction per video.

Both models are trained and evaluated on identical video-level splits with the same loss and metrics, so results are directly comparable.

Evaluation Framework

RESULTS NOT YET AVAILABLE

Accuracy
share of videos classified correctly

Precision
of videos flagged FAKE, how many are fake

Recall
of fake videos, how many are flagged

F1-score
harmonic mean of precision and recall

ROC-AUC
threshold-independent separability, where appropriate

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline CNN	—	—	—	—	—
Spatial + Frequency	—	—	—	—	—
Spatial + Temporal	—	—	—	—	—
Full Proposed Model	—	—	—	—	—

In-domain evaluation on the FaceForensics++ held-out test split, video-level predictions. Results will be populated after experimental evaluation.

Cross-dataset evaluation (if implemented)

Train	Test	Accuracy	Precision	Recall	F1	ROC-AUC
FaceForensics++	FaceForensics++ (in-domain)	—	—	—	—	—
FaceForensics++	Celeb-DF (cross-dataset, no fine-tuning)	—	—	—	—	—

Every reported number will be accompanied by the model checkpoint, data split and configuration used to produce it.

Ablation Study

PLANNED

Each component is added in turn so that its individual contribution can be measured on the same data split.

Exp.	Configuration	Spatial	Frequency	Temporal	Attention-based fusion	What it isolates
A	Spatial only	Yes	–	–	–	Value of the spatial branch alone (frame features averaged)
B	Spatial + Frequency	Yes	Yes	–	–	Contribution of frequency-domain features
C	Spatial + Temporal	Yes	–	Yes	–	Contribution of temporal modeling
D	Spatial + Frequency + Temporal	Yes	Yes	Yes	–	Complementarity of all three cues with simple aggregation
E	Full model with Transformer-based fusion	Yes	Yes	Yes	Yes	Added value of attention-based fusion over simple aggregation

Purpose

Determine whether each component actually contributes to performance, rather than assuming that a more complex model is better. Components that do not help will be reported as such.

Protocol

Identical video-level splits, training schedule and metrics for every configuration; results reported on the held-out test split. Configurations D and E differ only in how frame representations are aggregated.

Expected Contribution

EXPECTED OUTCOME · RESEARCH OBJECTIVES

The project aims to investigate whether hybrid feature learning can:

OBJECTIVE 1 Improve robustness to compression, quality variation and other nuisance factors	OBJECTIVE 2 Capture complementary forgery cues appearance, motion and spectral evidence in one model	OBJECTIVE 3 Reduce dependence on a single feature type so that failure of one cue does not fail the detector	OBJECTIVE 4 Improve generalization to different manipulation distributions, tested cross-dataset	OBJECTIVE 5 Provide a stronger research baseline with ablations that future work can build on
--	---	---	---	--

WHAT THIS PROJECT DOES NOT CLAIM

- It does not claim to “solve” deepfakes; detection is an evolving problem as generation methods change.
- It does not claim state-of-the-art performance; any such claim requires experimental evidence that does not yet exist.
- Improvements listed above are hypotheses. They will be confirmed or rejected by the planned experiments and ablations.

Limitations & Challenges

Computational requirements

Video preprocessing and sequence models are expensive; frame count, resolution and dataset size must be budgeted against available GPU time.

Compression and video quality variations

Re-encoding attenuates high-frequency artifacts and can invalidate cues learned on higher-quality data.

False positives / false negatives

Both error types carry real cost (wrongly discrediting genuine footage vs. missing a forgery); thresholds must be reported, not hidden.

Dependence on face detection / alignment

Detector failures, occlusion, profile views and multiple faces propagate errors into every downstream branch.

Dataset bias

Benchmarks share actors, capture set-ups and generators; a model may learn these regularities instead of forgery cues.

Generalization to unseen manipulation methods

New generators may leave different traces; cross-dataset testing probes but cannot guarantee transfer.

Highly sophisticated deepfakes

Some forgeries may leave few detectable traces in any of the three signals.

Future Scope

Directions beyond the current project scope; none are claimed as part of this work.

01

Audio-visual detection

Combine lip-sync and voice consistency with the visual branches.

02

Multimodal models

Joint reasoning over video, audio and metadata.

03

Real-time detection

Streaming inference with bounded latency per frame window.

04

Lightweight edge deployment

Compressed models for mobile and on-device screening.

05

Adversarial robustness

Resistance to perturbations designed to fool detectors.

06

Explainable AI

Attention and saliency maps to show which frames or regions drove a decision.

07

Emerging generation techniques

Diffusion-based and other new synthesis methods.

08

Larger cross-dataset evaluation

Broader benchmarks, including DFDC where compute allows.

Conclusion

- Deepfake detection is an important computer vision and cybersecurity problem.
- Existing detectors can perform well in-domain yet struggle to generalize to unseen data.
- The proposed system combines spatial, temporal and frequency-domain information.
- Transformer-based attention is used for sequence modeling and feature fusion.
- The project will evaluate, through ablations and cross-dataset tests, whether this hybrid architecture improves robustness.

“From detecting how a face looks, to how it moves, to the hidden patterns beneath it.”

References (1/2)

- [1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, “FaceForensics++: Learning to Detect Manipulated Facial Images,” in Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV), 2019.
- [2] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, “Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics,” in Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR), 2020.
- [3] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, “The DeepFake Detection Challenge (DFDC) Dataset,” arXiv:2006.07397, 2020.
- [4] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, “MesoNet: A Compact Facial Video Forgery Detection Network,” in Proc. IEEE Int. Workshop on Information Forensics and Security (WIFS), 2018.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), 2016.
- [6] A. Dosovitskiy et al., “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” in Proc. Int. Conf. on Learning Representations (ICLR), 2021.
- [7] A. Vaswani et al., “Attention Is All You Need,” in Advances in Neural Information Processing Systems (NeurIPS), 2017.
- [8] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, “Thinking in Frequency: Face Forgery Detection by Mining Frequency-Aware Clues,” in Proc. European Conf. on Computer Vision (ECCV), 2020.
- [9] R. Durall, M. Keuper, and J. Keuper, “Watch Your Up-Convolution: CNN Based Generative Deep Neural Networks Are Failing to Reproduce Spectral Distributions,” in Proc. IEEE/CVF CVPR, 2020.
- [10] J. Frank, T. Eule, A. Fischer, K. Rieck, D. Kolossa, and T. Holz, “Leveraging Frequency Analysis for Deep Fake Image Recognition,” in Proc. Int. Conf. on Machine Learning (ICML), 2020.

References (2/2)

- [11] E. Sabir, J. Cheng, A. Jaiswal, W. AbdAlmageed, I. Masi, and P. Natarajan, “Recurrent Convolutional Strategies for Face Manipulation Detection in Videos,” in Proc. IEEE/CVF CVPR Workshops, 2019.
- [12] D. Güera and E. J. Delp, “Deepfake Video Detection Using Recurrent Neural Networks,” in Proc. IEEE Int. Conf. on Advanced Video and Signal Based Surveillance (AVSS), 2018.
- [13] Y. Zheng, J. Bao, D. Chen, M. Zeng, and F. Wen, “Exploring Temporal Coherence for More General Video Face Forgery Detection,” in Proc. IEEE/CVF ICCV, 2021.
- [14] Y. Li, M.-C. Chang, and S. Lyu, “In Ictu Oculi: Exposing AI Created Fake Videos by Detecting Eye Blinking,” in Proc. IEEE WIFS, 2018.
- [15] L. Li, J. Bao, T. Zhang, H. Yang, D. Chen, F. Wen, and B. Guo, “Face X-ray for More General Face Forgery Detection,” in Proc. IEEE/CVF CVPR, 2020.
- [16] D. A. Coccomini, N. Messina, C. Gennaro, and F. Falchi, “Combining EfficientNet and Vision Transformers for Video Deepfake Detection,” in Proc. Int. Conf. on Image Analysis and Processing (ICIAP), 2022.
- [17] H. Zhao, W. Zhou, D. Chen, T. Wei, W. Zhang, and N. Yu, “Multi-Attentional Deepfake Detection,” in Proc. IEEE/CVF CVPR, 2021.
- [18] R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales, and J. Ortega-Garcia, “DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection,” Information Fusion, vol. 64, pp. 131–148, 2020.
- [19] L. Verdoliva, “Media Forensics and DeepFakes: An Overview,” IEEE Journal of Selected Topics in Signal Processing, vol. 14, no. 5, pp. 910–932, 2020.
- [20] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, “CNN-Generated Images Are Surprisingly Easy to Spot... For Now,” in Proc. IEEE/CVF CVPR, 2020.

Citation style: IEEE. Bracketed numbers are used on the Related Work, Temporal Modeling and Dataset slides.

Questions & Discussion

Thank you

Deepfake Detection via Hybrid Spatial-Temporal-Frequency Feature Learning with Transformer-Based Fusion

Ayush Tiwari · Ayushman Patel · Atharv Shukla · Bhanupratap Chourasia

